Response to Reviewers' Comments

We sincerely thank the Associate Editor and all reviewers for their helpful comments and suggestions. We have revised the manuscript and included here a point-by-point response. The main changes are the following:

1) ...
2) ...

Major changes are highlighted in blue in our revised manuscript. The references like equation numbers refer to those in the manuscript, except for those prefixed by "R".

Response to Reviewer 1

Recommendation: Major Revision

Comment 1.1. *A lot of comments.*

Response 1.1. Thank you for your comment. Please always use the formal "thank you" instead of "thanks". Please always start your response politely and in a conciliatory tone even if you are frustrated with the comment. We can be firm, friendly and respectful. After all, the reviewer has sacrificed his or her precious time to read our manuscript and provide comments, which he or she is not obligated to do.

Using the xr package, I can reference Section I, Fig. 2a and (1) of the main paper tips.tex. To do this, you need to build response.tex in the same folder containing tips.aux.

Next, I cite [13], [25], which are in the main paper. The references displayed are from tips.aux and tips.bbl thanks to xr. Finally, the catchfilebetweentags package allows me to quote the following from Section IX in tips.tex:

> This ensures that whatever changes you make in the tex file are reflected in the response. Examples given below are used in response.tex.

Note that the tag name cannot include underscores. Another quote from tips.tex:

> Other examples are as follows:
>
> $$\Lambda_{\mathrm{SGLR}}(k,t) = \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, \tag{3}$$
> $$\tau_{\mathrm{SGLR}}(b) = \min\{\tau(b), \tau_n(b)\}, \tag{4}$$
> $$\tau_{\mathrm{W-SGLR}}(b) = \min\{\widetilde{\tau}(b), \widetilde{\tau_n}(b)\}, \tag{5}$$
> $$\Phi_i = \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \dots, N,$$
> $$\widetilde{\mathbf{C}} = \mathrm{one\_hot}\left(\widetilde{\mathbf{L}}\right), \tag{6}$$
> $$\mathbf{x} = \mathbf{B}^{\mathsf{H}}\mathbf{\Sigma}\mathbf{M}^{\mathsf{T}}\mathbf{\Phi}\mathbf{M}\mathbf{A}^{\mathsf{T}},$$
> $$d = D(p_X \,\|\, p_Z).$$

The equation numbers shown here are the same as those in the main paper, which is great!

Comment 1.2. *Do something.*

Response 1.2. Thank you for your suggestion. I will now cite some additional references that appear only in the bibliography for this response using [A1]–[A3] and [A4, Thm 1]. Note that we use `\citeA` to cite these references, while `\cite` will cite references from the main paper.

Response to Reviewer 2

Recommendation: Major Revision

Comment 2.1. *Why does this hold?*

Response 2.1. Thank you for your question. Please refer to the following equation:

$$f(x) = \sum_{i>0} y_i. \tag{R1}$$

Note the prefix "R" in the equation that is in this response. From (R1) and (3) in the main paper, the result follows immediately.

Comment 2.2. *A lot of comments.*

Response 2.2. Thank you for your comment. Repeated quote from tips.tex:

Other examples are as follows:

$$\Lambda_{\mathrm{SGLR}}(k,t) = \max\left\{\Lambda(k,t), \Lambda_n(k,t)\right\}, \tag{3}$$

$$\tau_{\mathrm{SGLR}}(b) = \min\{\tau(b), \tau_n(b)\}, \tag{4}$$

$$\tau_{\mathrm{W-SGLR}}(b) = \min\{\widetilde{\tau}(b), \widetilde{\tau_n}(b)\}, \tag{5}$$

$$\Phi_i = \tau_{\mathrm{W-SGLR}}(b) + i^2, \text{ for } i = 1, \ldots, N,$$

$$\widetilde{\mathbf{C}} = \mathrm{one\_hot}\left(\widetilde{\mathbf{L}}\right), \tag{6}$$

$$\mathbf{x} = \mathbf{B}^{\mathsf{H}}\mathbf{\Sigma}\mathbf{M}^{\mathsf{T}}\mathbf{\Phi}\mathbf{M}\mathbf{A}^{\mathsf{T}},$$

$$d = D(p_X \,\|\, p_Z).$$

We remark that it can be shown that

$$\Lambda(k) = \int_{\Omega} \tau(b)\, \mathrm{d}\nu(b), \tag{R2}$$

$$\Gamma(k) = \int_{\mathcal{A}} f(x)\, \mathrm{d}\mu(x). \tag{R3}$$

The equation numbers shown in the box are the same as those in the main paper, which is great! The command `\revision` takes care of this. But it has limitations. See the following examples.

Comment 2.3. *Some other comments.*

Response 2.3. Thank you for your comment. Please refer to Fig. 2. We provide the following example, reproduced here:

Example 2. *Consider the subset of vertices V_0 of an unweighted directed graph G. Suppose that we consider the adjacency matrix $\mathbf{A}_G$ as the graph shift operator for G. The adjacency matrix $\mathbf{A}_{H_0}$ of the subgraph H_0 shown is a submatrix consisting of the even-indexed rows and columns of $\mathbf{A}_G^2$. Any filter shift invariant with respect to (w.r.t.) $\mathbf{A}_{H_0}$ can be viewed as a polynomial of $\mathbf{A}_G^2$ composed with a projection to the vertices of H_0. See [2] for more details.*

Kernel-based techniques have more flexibility in filtering and reconstruction, since they introduce nonlinearity and generalize existing approaches. The graph signal is modeled as a random nonlinear function of an arbitrary input with a specific covariance structure adapted to the graph. The covariance structure is based on a scalar-valued kernel for differentiating the inputs and the graph structure for regularizing the smoothness of the random graph signal.

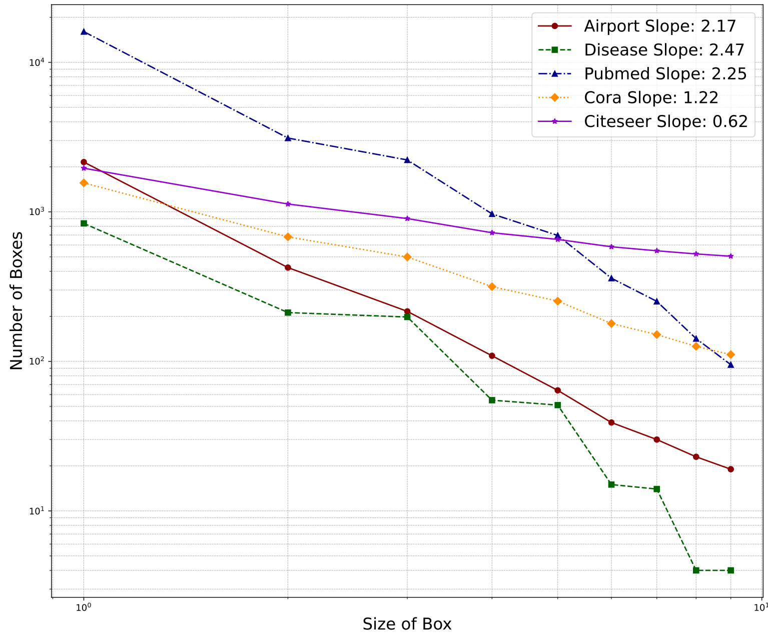

Fig. 3. The fractal dim of datasets used in our experiments. We use the Compact-Box-Burning algorithm to compute the log-log slope (fractal dim) of the box size and the minimum number of boxes needed to cover the graph.

Note the use of `\xrcounterref` in the macro `\revision` to set the numbering of environments correctly. **If you know how to make this automatic, let me know.**

The macro `\revision` takes in an optional list of pairs of environment name and label, separated by semicolons ;. Each pair is of the form `environment_name>label`. The environment name can be figure, table, Theorem, Lemma, Example, etc. The label is the label of the environment in the main paper. For example, in the above example, we have `Example>eg:subGSP;figure>fig:fractal`.

Comment 2.4. *Some other comments.*

Response 2.4. We provide the following theorem, reproduced here:

The following is another theorem.

Theorem 2. *The limit posterior covariance of $f(v_0, \mathbf{t})$ over $\mathcal{T}_0$ converges:*

$$\lim_{M_0\to\infty} \int_{\mathcal{T}_0} \mathrm{var}(f(v_0, \mathbf{t}) \,|\, \mathbf{y}(M_0)) \, \mathrm{d}\tau(\mathbf{t}) = \int_{\mathcal{T}_0} \mathrm{var}(f(v_0, \mathbf{t}) \,|\, \mathbf{z}) \, \mathrm{d}\tau(\mathbf{t}). \tag{9}$$

Proof: See Appendix A. ■

Corollary 1. *From Theorem 2, we have that the limit posterior covariance of $f(v_0, \mathbf{t})$ over $\mathcal{T}_0$ is equal to the posterior covariance of $f(v_0, \mathbf{t})$ over $\mathcal{T}_S$:*

$$\int_{\mathcal{T}_0} \mathrm{var}(f(v_0, \mathbf{t}) \,|\, \mathbf{y}(\infty)) \, \mathrm{d}\tau(\mathbf{t}) = \int_{\mathcal{T}_0} \mathrm{var}(f(v_0, \mathbf{t}) \,|\, \mathbf{z}) \, \mathrm{d}\tau(\mathbf{t}). \tag{10}$$

Note that the theorem and corollary numbers shown in the box are the same as those in the main paper.

Comment 2.5. *Some other comments.*

Response 2.5. Thank you for your comment. Blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah blah.

In the macro `\revision`, floats like tables and figures have their placement options replaced with "H". For example, the following figure and tables from the main text are reproduced here:

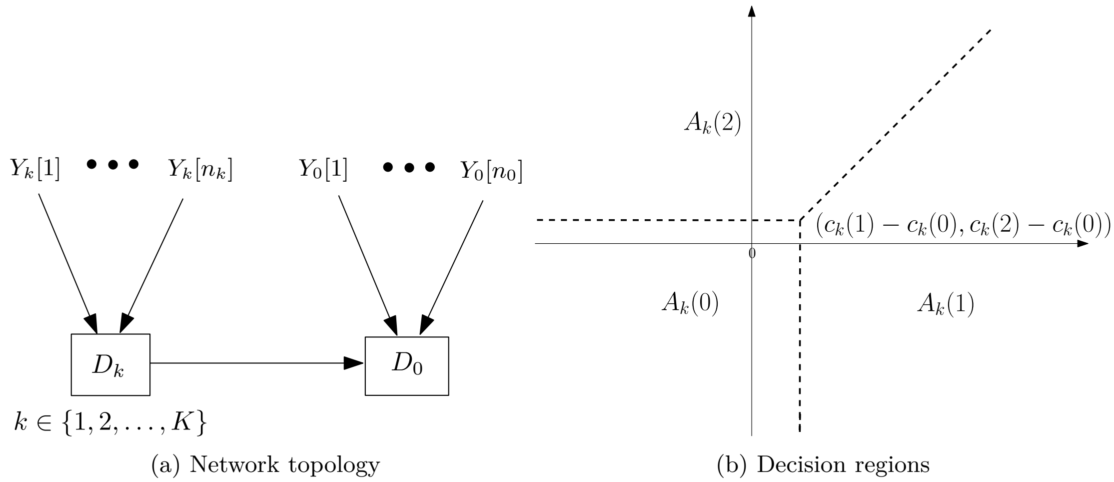

Fig. 2. A tandem network and its decision regions.

TABLE II
COMPARISON WITH BASE MODELS ON HOMOPHILIC DATASETS

Dataset	F-GRAND	L-F-GRAND	P-F-GRAND	R-F-GRAND
Cora	84.68±1.01	84.73±1.12	**84.76±1.10**	**85.16±1.18**
Citeseer	73.71±1.00	**73.75±1.64**	73.72±1.67	**74.05±1.60**
Pubmed	79.14±1.68	**79.29±2.09**	79.19±1.55	**79.38±1.59**
Photo	92.54±1.36	92.55±1.26	**92.70±1.31**	**93.16±1.33**
CS	92.93±0.25	**93.02±0.24**	92.98±0.21	**93.37±0.25**

Comment 2.6. *Some other comments.*

Response 2.6. Unfortunately, the catchfilebetweentags package does not support nested tags, so we have to use the starred version of `\revision` (using the clipboard package) to copy and paste the parts of the latex content we want. See the following example:

Kernel-based techniques have more flexibility in filtering and reconstruction, since they introduce nonlinearity and generalize existing approaches. The graph signal is modeled as a random nonlinear function of an arbitrary input with a specific covariance structure adapted to the graph. The covariance structure is based on a scalar-valued kernel for differentiating the inputs and the graph structure for regularizing the smoothness of the random graph signal.

Additional References

[A1] D. Acemoglu, M. A. Dahleh, I. Lobel, and A. Ozdaglar, "Bayesian learning in social networks," *Review of Economic Studies*, vol. 78, no. 4, pp. 1201–1236, Mar. 2011.

[A2] P.-N. Chen and A. Papamarcou, "Error bounds for parallel distributed detection under the Neyman-Pearson criterion," in *Proc. IEEE Int. Symp. Inform. Theory*, Mar. 1995, pp. 528–533.

[A3] T. M. Cover, "Hypothesis testing with finite statistics," *Annals of Mathematical Statistics*, vol. 40, no. 3, pp. 828–835, 1969.

[A4] X. Jian, W. P. Tay, and Y. Eldar, "Kernel based reconstruction for generalized graph signal processing," *IEEE Trans. Signal Process.*, vol. 72, pp. 2308 – 2322, 2024.